OrchardScout: exploring an orchard of unknown size with one camera

Dhruv Sharma · proof of concept, November 2026 · code, video and every run: **github.com/dhruuvsharma/OrchardScout**

Task

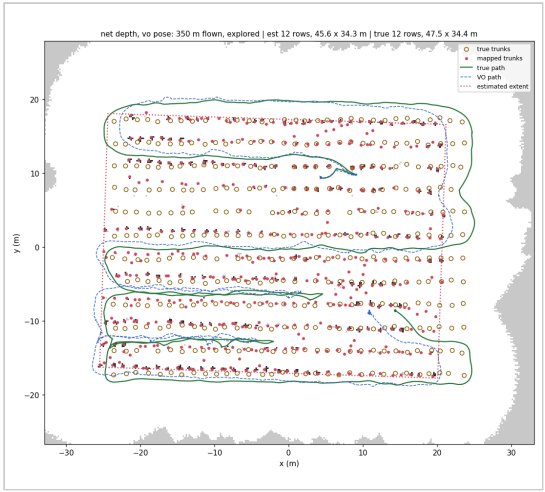
A drone flies under the canopy of a simulated orchard. Using **only an RGB camera** it must localise itself, explore, build a map and work out **how big the orchard is** (rows, row length, spacing, extent), which it is never told. Every run draws a new orchard (4–20 rows, 20–120 m long, random row and tree spacing, missing trees, jitter) and starts the drone in an alley or outside an edge, facing anywhere. Simulator ground truth only scores the run. One stated prior: the flight controller holds a known height (1.0 m), which fixes monocular scale. No IMU, GPS or depth sensor.

Method

- **Depth:** Depth Anything V2 (metric) is ~5× off in scale at drone height; each frame it is rescaled so a RANSAC ground plane (bottom 12 % of the image) sits 1.0 m below the camera. Sky, depth-edge ramps and "rising" open-field ground are masked.
- **Visual odometry:** KLT tracks + PnP/RANSAC on keyframe depth; the ground plane fixes roll, pitch and height every frame.
- **Mapping:** occupancy grid (free space along every ray) and trunk landmarks (vertical point clusters within 5 m, corrected for seeing only the trunk's front).
- **Drift correction from orchard structure:** a *row compass* (rows are parallel: RANSAC lines of trees hold the heading), a *row lattice* (rows are evenly spaced: the sideways position is corrected modulo the spacing) and landmark re-sighting.
- **Size:** rows = density peaks across the row direction; spacings from trunks seen within 4 m (accurate depth) over a few seconds; width = (rows−1)×spacing.
- **Exploration:** frontier-based (Dijkstra on an inflated grid). Only frontiers near mapped trees count, so open field beyond the edges is ignored and the run **stops by itself** once the orchard has been seen. Paths prefer alleys (depth is poor over open grass); a safety layer keeps clear of trunks.

What I learned

- Monocular depth fails in structured ways: wrong global scale, compression with distance, open grass bent upwards. Masks, range limits and close-range measurements beat trying to "correct" it.
- Most drift comes from turns at the orchard edge, facing open field. The orchard's own geometry (parallel, evenly spaced rows) is a strong, cheap prior that removes most of it.
- A noisy correction can be worse than none: a heading fitted from a few trees random-walked; robust line fits fixed it.



Camera-only run, medium orchard #0 (12 rows, 47.5 x 34.4 m) against ground truth: true path (green), odometry (blue), mapped trunks (red), true trunks (circles), estimate (dotted): 12 rows, 45.6 x 34.3 m.

Results: autonomous runs, each on a new orchard

Final code. **Held-out**: ten orchards (two rounds of five) never run before the code was frozen; an earlier held-out round found two failures, since fixed, and its orchards joined the development set. **Camera**: the real system. **Perfect**: simulator depth and pose (upper bound). Red: wrong row count.

set	orchard (true size)	start	rows camera	length camera	width camera	flown camera	coverage camera	collisions camera	rows perfect	length perfect	width perfect
held-out	small #5: 6 rows, 32 x 23 m	alley	6 / 6	+9.4 %	-4.1 %	158 m	100 %	0	6	-0.1 %	+0.5 %
held-out	small #6: 5 rows, 24 x 16 m	edge	5 / 5	+1.4 %	-5.8 %	92 m	100 %	0	5	+0.2 %	-2.0 %
held-out	medium #4: 11 rows, 55 x 59 m	alley	11 / 11	+2.1 %	-9.4 %	695 m	100 %	0	11	-0.0 %	-0.4 %
held-out	medium #5: 11 rows, 64 x 45 m	edge	11 / 11	+0.9 %	-2.9 %	614 m	100 %	0	11	-0.0 %	-0.3 %
held-out	large #2: 19 rows, 92 x 98 m	edge	18 / 19	+3.3 %	-12.1 %	1639 m	99 %	0	19	+0.0 %	-0.9 %
held-out	small #7: 6 rows, 33 x 27 m	alley	6 / 6	+2.4 %	-8.9 %	201 m	100 %	0	6	+0.2 %	+0.1 %
held-out	small #8: 6 rows, 34 x 20 m	edge	6 / 6	-0.8 %	-8.6 %	134 m	100 %	0	6	-0.0 %	-1.1 %
held-out	medium #6: 10 rows, 50 x 37 m	alley	10 / 10	-0.1 %	-9.4 %	345 m	100 %	0	10	+0.0 %	-0.9 %
held-out	medium #7: 12 rows, 66 x 59 m	edge	13 / 12	+3.1 %	+1.5 %	762 m	99 %	0	12	+0.0 %	-0.8 %
held-out	large #3: 19 rows, 89 x 97 m	alley	22 / 19	+4.6 %	+7.2 %	1449 m	100 %	0	19	+0.0 %	-0.3 %
development	small #0: 6 rows, 23 x 16 m	alley	6 / 6	+7.4 %	-0.1 %	113 m	100 %	0	6	+0.0 %	-2.7 %
development	small #1: 5 rows, 32 x 14 m	edge	5 / 5	+0.4 %	-2.3 %	122 m	100 %	0	5	-41.3 %	-1.9 %
development	small #2: 6 rows, 24 x 27 m	alley	6 / 6	+2.6 %	-8.4 %	205 m	100 %	0	6	+0.1 %	-1.0 %
development	small #3: 6 rows, 21 x 27 m	alley	7 / 6	+16.3 %	+6.6 %	150 m	100 %	0	6	+0.3 %	-0.3 %
development	small #4: 6 rows, 28 x 30 m	edge	6 / 6	+13.4 %	-9.6 %	268 m	100 %	0	6	-0.1 %	+0.1 %
development	medium #0: 12 rows, 47 x 34 m	alley	12 / 12	-4.0 %	-0.0 %	350 m	100 %	0	12	+0.0 %	-0.9 %
development	medium #1: 10 rows, 68 x 31 m	edge	10 / 10	+0.8 %	-1.9 %	341 m	100 %	0	10	+0.0 %	+0.1 %
development	medium #2: 12 rows, 49 x 60 m	edge	12 / 12	+12.2 %	-9.4 %	509 m	100 %	0	12	+0.1 %	-1.4 %
development	medium #3: 12 rows, 45 x 59 m	alley	13 / 12	+14.9 %	-2.4 %	435 m	100 %	0	12	-0.1 %	-1.1 %
development	large #0: 19 rows, 90 x 56 m	alley	19 / 19	+0.8 %	-1.3 %	1041 m	100 %	100	19	+0.0 %	-0.6 %
development	large #1: 17 rows, 118 x 55 m	edge	19 / 17	+0.1 %	+10.7 %	884 m	100 %	0	17	+0.0 %	+0.1 %

Collisions: 0.1 s steps in contact with a tree. Large #0 camera: second attempt (the first livelocked after 585 m). Every run and map: repository README.

Limitations and next steps

- Simulation only (MuJoCo; Isaac Sim needs an RTX GPU). Next: Isaac Lab / PX4 and real orchard footage.
- **Failures, camera only:** on the largest orchards (0.9–1.6 km of flight) drift near the edges passed half a row spacing and added or dropped edge rows (large #1, #2, #3): no loop closure yet, and re-seen rows are the natural constraint. A 7–11 % short spacing in wide alleys added a row (small #3, medium #3, medium #7). One run livelocked beside a trunk (now ended by a watchdog); its re-run touched a tree for 10 s. **Perfect sensing:** one run stopped early (free space is mapped to 8 m, trunks to 5 m). A revised explorer, run on all 21 orchards afterwards, had none of these exploration failures in 42 runs and the same camera-only row count (15 of 21): the limit now is drift and depth, not exploration.
- Exploration is classical. Next: a learned exploration policy against this baseline, and an adversarial "environment agent" that generates orchards (lighting, fog, irregular rows, slopes) to find where camera-only navigation breaks.